ARYAN MEENA

Data Scientist | Python | SQL | Machine Learning | PySpark

(617) 861-1307 | araj7042@gmail.com | LinkedIn | GitHub | Care Dashboard

SKILLS

Python, SQL, machine learning, scikit-learn, XGBoost, LightGBM, PyTorch, PySpark, statistical modeling, hypothesis testing, A/B testing, regression, classification, clustering, time series forecasting, feature engineering, cross-validation, model evaluation, bootstrap, pandas, BigQuery, AWS, Tableau, Git

EXPERIENCE

Boston University | ***Data Scientist (Research Assistantship)*** Sep 2025 – May 2026

- Forecast 5-day currency volatility out of sample at R^2 0.25 vs a 0.05 noise ceiling across 8 CME currency futures, testing 6 models under purged walk-forward validation rerun 20 times on rotated targets.
- Showed a 7-variable regression improved on the standard HAR volatility benchmark more than a 42-feature random forest did (Diebold-Mariano t -4.95 vs -3.98), so the simpler model was the one to deploy.
- Engineered 42 non-redundant features across 8 blocks for CME currency futures, cutting the feature matrix's condition number from 1.4M to 141 before modeling with purged walk-forward validation.
- Caught 5 feature-engineering defects before any model was fitted, including a pandas dtype bug that would have leaked 615 constructed zeros into the targets and given models an unearned accuracy score.
- Flagged next-day two-sigma currency moves at ROC AUC 0.64 vs a 0.51 noise ceiling.
- Traced a 15-year trading signal to biased opening prices in a free data feed (Sharpe -1.36 on vendor data vs +0.25 on exchange data), protecting the research from a false result; paper **published in Risks (MDPI)**.

Boston University | ***Graduate Teaching Assistant*** Jan 2026 – May 2026

- Explained statistics and model results in plain language to 60+ graduate students in weekly office hours for Advanced Machine Learning and Data Science with Python, turning debugging problems into clear next steps.

PROJECTS

Hospital Readmission Risk | Python, LightGBM, XGBoost, Tableau Care Dashboard

- Applied machine learning to 101,766 hospital encounters, beating a published benchmark (AUC 0.689 vs 0.664) with LightGBM over XGBoost and logistic regression under patient-grouped validation.
- Ranked patients by risk so the top 5% captured 1,696 readmissions vs 566 at random (**3.0x lift**).
- Audited the model by age, race and sex, confirmed no race or sex gap, and recommended age-specific thresholds.

WikiFlow: Editor Dropout at Scale | PySpark, GCP Dataproc, BigQuery Springer Paper

- Processed 60M Wikipedia edit events (~25 GB) on a 5-node Spark cluster on Google Cloud to predict which of 1.16M new editors will drop out, in about 2 hours for roughly $2.50; paper published by Springer.
- Reached AUC 0.909 with a distributed neural network, 6.4 points above logistic regression.

Service Operations Analytics | Python, AWS, Forecasting Live Dashboard

- Predicted NYC 311 service demand 14 days ahead for 5 boroughs, refreshed daily with no manual steps.
- Automated a daily AWS pipeline over 130K+ records that publishes only after 17 data checks pass, at about $0.03 per run, with alarms for broken or stale data so reports stay trustworthy.

EDUCATION

MS, Applied Data Analytics, Boston University (GPA 3.75/4.0) May 2026

- Led a six-person team at the BU MedAI Hackathon to a **podium finish** among 40+ teams; 5 first-author papers.
- Coursework: Data Mining, Data Analytics and Visualization, Foundations of Machine Learning, Big Data Analytics.

BTech, Electronics and Communication Engineering, Indian Institute of Space Science and Technology May 2023

- Coursework: Statistics and Probability, Computer Vision, Machine Learning for Signal Processing.